

Лабораторная работа № 6. Эмуляция и измерение потерь пакетов в глобальных сетях

Абакумова О. М.

Российский университет дружбы народов, Москва, Россия

Информация

- Абакумова Олеся Максимовна
- Студентка
- Российский университет дружбы народов
- 1132220832@pfur.ru
- <https://github.com/omabakumova>



Цель работы

Основной целью работы является знакомство с принципами работы дисциплины очереди Token Bucket Filter, которая формирует входящий/исходящий трафик для ограничения пропускной способности, а также получение навыков моделирования и исследования поведения трафика посредством проведения интерактивного и воспроизводимого экспериментов в Mininet

Предварительные сведения

Token Bucket Filter (TBF) представляет собой алгоритм, используемый в сетях с коммутацией пакетов для ограничения пропускной способности и пиковой нагрузки трафика. Передача поступающих в очередь системы (queue) пакетов данных осуществляется при наличии в специальном буфере (bucket) необходимого числа разрешений на передачу (или токенов). Токены могут быть представлены в виде пакетов или числа байтов, поступающих в буфер (корзину) фиксированного размера с фиксированной скоростью.

Задание

1. Задайте топологию, состоящую из двух хостов и двух коммутаторов с назначенной по умолчанию mininet сетью 10.0.0.0/8.
2. Проведите интерактивные эксперименты по ограничению пропускной способности сети с помощью TBF в эмулируемой глобальной сети.
3. Самостоятельно реализуйте воспроизводимые эксперимент по применению TBF для ограничения пропускной способности. Постройте соответствующие графики.

Выполнение лабораторной работы

Запуск лабораторной топологии

```
mininet@192.168.56.101's password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-42-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

Last login: Sat Nov  8 08:11:44 2025 from 192.168.56.1
mininet@mininet-vm:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 192.168.56.101  netmask 255.255.255.0  broadcast 192.168.56.255
    ether 08:00:27:40:a4:3f  txqueuelen 1000  (Ethernet)
    RX packets 117  bytes 18812 (18.8 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 53  bytes 8991 (8.9 KB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
    ether 08:00:27:19:7d:07  txqueuelen 1000  (Ethernet)
    RX packets 374  bytes 45027 (45.0 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 394  bytes 35962 (35.9 KB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    loop txqueuelen 1000  (Local Loopback)
    RX packets 56  bytes 4984 (4.9 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 56  bytes 4984 (4.9 KB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

mininet@mininet-vm:~$
```

Рис. 1: Запуск виртуальной машины

Запуск лабораторной топологии

```
mininet@mininet-vm:~$ xauth list $DISPLAY
mininet-vm/unix:10 MIT-MAGIC-COOKIE-1 c513ec3fcc72804731e9f74fa948aa4e
mininet@mininet-vm:~$ sudo -i
root@mininet-vm:~# xauth add mininet-vm/unix:10 MIT-MAGIC-COOKIE-1 c513ec3fcc72804731e9f74fa948aa4e
root@mininet-vm:~# logout
mininet@mininet-vm:~$
```

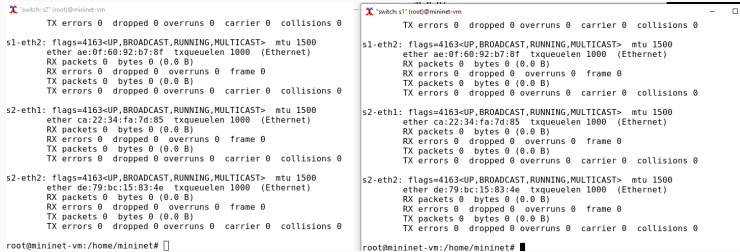
Рис. 2: Исправление прав запуска X-соединения

Запуск лабораторной топологии



Рис. 3: Задание топологии

Запуск лабораторной топологии



The image displays two terminal windows side-by-side, showing network statistics for switches s1 and s2. The left window is titled "switch: s2" and the right window is titled "switch: s1". Both windows show the same set of statistics for their respective switches.

```
switch: s2 (root)@mininet-vm
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

s1-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
ether ae:0f:60:92:b7:8f txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

s2-eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
ether ca:22:34:fa:7d:85 txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

s2-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
ether de:79:bc:15:83:4e txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet#
```

```
switch: s1 (root)@mininet-vm
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

s1-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
ether ae:0f:60:92:b7:8f txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

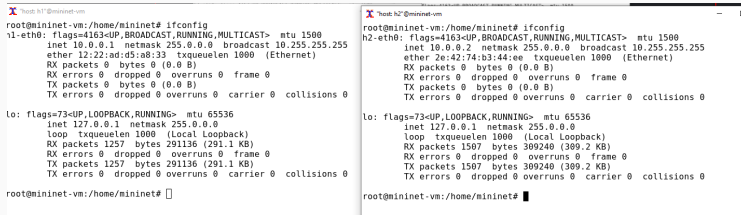
s2-eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
ether ca:22:34:fa:7d:85 txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

s2-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
ether de:79:bc:15:83:4e txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet#
```

Рис. 4: Информация на коммутаторах s1 и s2

Запуск лабораторной топологии



The image shows two terminal windows side-by-side, both titled "host: h1" and "host: h2" respectively, indicating they are running on hosts h1 and h2 in a Mininet environment. Both windows show the output of the 'ifconfig' command for the root user at the /home/mininet# prompt.

Terminal h1:

```
root@mininet-vm:/home/mininet# ifconfig
v1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.1 netmask 255.0.0.0 broadcast 10.255.255.255
    ether 12:22:ad:d5:a8:33 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1257 bytes 291136 (291.1 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1257 bytes 291136 (291.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet#
```

Terminal h2:

```
root@mininet-vm:/home/mininet# ifconfig
h2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.2 netmask 255.0.0.0 broadcast 10.255.255.255
    ether 2e:42:74:b3:44:ee txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1507 bytes 309240 (309.2 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1507 bytes 309240 (309.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet#
```

Рис. 5: Информация на хостах h1 и h2

Запуск лабораторной топологии

```
root@host: h1" @mininet-vn
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
loop txqueuelen 1000 (Local Loopback)
RX packets 1257 bytes 291136 (291.1 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 1257 bytes 291136 (291.1 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 4
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=4.78 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.336 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.066 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3028ms
rtt min/avg/max/mdev = 0.062/1.310/4.777/2.004 ms
root@mininet-vm:/home/mininet#
```

```
root@host: h2" @mininet-vn
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

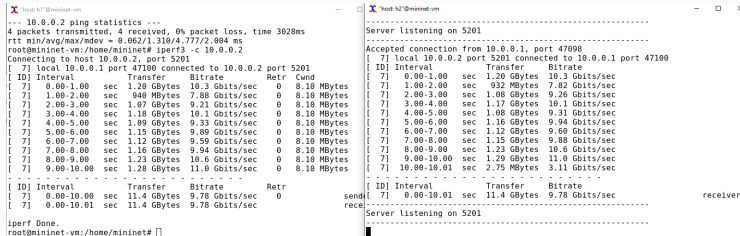
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
loop txqueuelen 1000 (Local Loopback)
RX packets 1507 bytes 309240 (309.2 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 1507 bytes 309240 (309.2 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# ping 10.0.0.1 -c 4
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data:
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=2.66 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0.060 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0.074 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0.061 ms

--- 10.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3052ms
rtt min/avg/max/mdev = 0.060/0.713/2.658/1.122 ms
root@mininet-vm:/home/mininet#
```

Рис. 6: Проверка подключения между хостами

Запуск лабораторной топологии



```
host: h1@mininet-vm
--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3028ms
rtt min/avg/max/ndev = 0.062/1.310/4.777/2.004 ms
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 47100 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate      Retr  Cwnd
[ 7] 0.00-1.00 sec  1.20 GBytes 10.3 Gbits/sec  0    8.10 MBytes
[ 7] 1.00-2.00 sec  940 MBytes 7.88 Gbits/sec  0    8.10 MBytes
[ 7] 2.00-3.00 sec  1.07 GBytes 9.21 Gbits/sec  0    8.10 MBytes
[ 7] 3.00-4.00 sec  1.18 GBytes 10.1 Gbits/sec  0    8.10 MBytes
[ 7] 4.00-5.00 sec  1.09 GBytes 9.33 Gbits/sec  0    8.10 MBytes
[ 7] 5.00-6.00 sec  1.15 GBytes 9.89 Gbits/sec  0    8.10 MBytes
[ 7] 6.00-7.00 sec  1.12 GBytes 9.59 Gbits/sec  0    8.10 MBytes
[ 7] 7.00-8.00 sec  1.16 GBytes 9.94 Gbits/sec  0    8.10 MBytes
[ 7] 8.00-9.00 sec  1.23 GBytes 10.6 Gbits/sec  0    8.10 MBytes
[ 7] 9.00-10.00 sec 1.28 GBytes 11.0 Gbits/sec  0    8.10 MBytes
-----
[ ID] Interval      Transfer    Bitrate      Retr
[ 7] 0.00-10.00 sec 11.4 GBytes 9.78 Gbits/sec  0
[ 7] 0.00-10.01 sec 11.4 GBytes 9.78 Gbits/sec
iperf Done.
root@mininet-vm:/home/mininet#
```

```
host: h2@mininet-vm
Server listening on 5201
Accepted connection from 10.0.0.1, port 47098
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 47100
[ ID] Interval      Transfer    Bitrate
[ 7] 0.00-1.00 sec  1.20 GBytes 10.3 Gbits/sec
[ 7] 1.00-2.00 sec  932 MBytes 7.82 Gbits/sec
[ 7] 2.00-3.00 sec  1.08 GBytes 9.26 Gbits/sec
[ 7] 3.00-4.00 sec  1.17 GBytes 10.1 Gbits/sec
[ 7] 4.00-5.00 sec  1.08 GBytes 9.31 Gbits/sec
[ 7] 5.00-6.00 sec  1.16 GBytes 9.94 Gbits/sec
[ 7] 6.00-7.00 sec  1.12 GBytes 9.60 Gbits/sec
[ 7] 7.00-8.00 sec  1.15 GBytes 9.88 Gbits/sec
[ 7] 8.00-9.00 sec  1.23 GBytes 10.6 Gbits/sec
[ 7] 9.00-10.00 sec 1.29 GBytes 11.0 Gbits/sec
[ 7] 10.00-10.01 sec 2.75 MBytes 3.11 Gbits/sec
-----
[ ID] Interval      Transfer    Bitrate
[ 7] 0.00-10.01 sec 11.4 GBytes 9.78 Gbits/sec
Server listening on 5201
-----
```

Рис. 7: Запуск iPerf3 на обоих хостах в разных режимах на обоих хостах

Команду `tc` можно применить к сетевому интерфейсу устройства для формирования исходящего трафика. Требуется ограничить скорость отправки данных с конечного хоста с помощью фильтра Token Bucket Filter (tbf)

Интерактивные эксперименты

```
"host: h1"@mininet-vm
rtt min/avg/max/mdev = 0.062/1.310/4.777/2.004 ms
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 47100 connected to 10.0.0.2 port 5201
[ ID] Interval           Transfer     Bitrate        Retr    Cwnd
[ 7]  0.00-1.00    sec   1.20 GBytes  10.3 Gbits/sec    0     8.10 MBytes
[ 7]  1.00-2.00    sec   940 MBytes  7.88 Gbits/sec    0     8.10 MBytes
[ 7]  2.00-3.00    sec   1.07 GBytes  9.21 Gbits/sec    0     8.10 MBytes
[ 7]  3.00-4.00    sec   1.18 GBytes  10.1 Gbits/sec    0     8.10 MBytes
[ 7]  4.00-5.00    sec   1.09 GBytes  9.33 Gbits/sec    0     8.10 MBytes
[ 7]  5.00-6.00    sec   1.15 GBytes  9.89 Gbits/sec    0     8.10 MBytes
[ 7]  6.00-7.00    sec   1.12 GBytes  9.59 Gbits/sec    0     8.10 MBytes
[ 7]  7.00-8.00    sec   1.16 GBytes  9.94 Gbits/sec    0     8.10 MBytes
[ 7]  8.00-9.00    sec   1.23 GBytes  10.6 Gbits/sec    0     8.10 MBytes
[ 7]  9.00-10.00   sec   1.28 GBytes  11.0 Gbits/sec    0     8.10 MBytes
- - - - -
[ ID] Interval           Transfer     Bitrate        Retr
[ 7]  0.00-10.00   sec   11.4 GBytes  9.78 Gbits/sec    0
[ 7]  0.00-10.01   sec   11.4 GBytes  9.78 Gbits/sec
sender
receiver

iperf Done.
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root tbf rate 10gb
it burst 5000000 limit 15000000
root@mininet-vm:/home/mininet#
```

Рис. 8: Установка пропускной способности на 10 Гбит/с на интерфейсе h1-eth0

```
root@mininet-vm:/home/mininet# egrep '^CONFIG_HZ_[0-9]+' /boot/config-`uname -r`  
CONFIG_HZ_250=y  
root@mininet-vm:/home/mininet#
```

Рис. 9: Извлечение тактовой частоты

Для расчёта значения всплеска (burst) необходимо скорость передачи (10 Гбит/с или 10 Gbps = 10,000,000,000 bps) разделить на полученное таким образом значение HZ (на хосте h1 HZ = 250): $\text{Burst} = 10,000,000,000 / 250 = 40,000,000 \text{ bits} = 40,000,000 / 8 \text{ bytes} = 5,000,000 \text{ bytes}$.

Интерактивные эксперименты

```
root@mininet-vm:/home/mininet# egrep '^CONFIG_HZ_[0-9]+' /boot/config-`uname -r`
CONFIG_HZ_250=y
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 47184 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate    Retr  Cwnd
[ 7] 0.00-1.00 sec  1.09 GBytes  9.36 Gbits/sec  0  8.23 MBytes
[ 7] 1.00-2.00 sec  1.11 GBytes  9.56 Gbits/sec  0  8.23 MBytes
[ 7] 2.00-3.00 sec  1.10 GBytes  9.49 Gbits/sec  0  8.23 MBytes
[ 7] 3.00-4.00 sec  1.11 GBytes  9.53 Gbits/sec  0  8.23 MBytes
[ 7] 4.00-5.00 sec  1.11 GBytes  9.54 Gbits/sec  0  8.23 MBytes
[ 7] 5.00-6.00 sec  1.12 GBytes  9.61 Gbits/sec  0  8.23 MBytes
[ 7] 6.00-7.00 sec  1.10 GBytes  9.49 Gbits/sec  0  8.23 MBytes
[ 7] 7.00-8.00 sec  1.09 GBytes  9.40 Gbits/sec  0  8.23 MBytes
[ 7] 8.00-9.00 sec  1.12 GBytes  9.58 Gbits/sec  0  8.23 MBytes
[ 7] 9.00-10.00 sec 1.11 GBytes  9.50 Gbits/sec  0  8.23 MBytes
[ ID] Interval      Transfer    Bitrate    Retr  Cwnd
[ 7] 0.00-10.00 sec 11.1 GBytes  9.51 Gbits/sec  0
[ 7] 0.00-10.02 sec 11.1 GBytes  9.49 Gbits/sec  0
sender
receiver
iperf Done.
root@mininet-vm:/home/mininet#
```

```
Server listening on 5201
Accepted connection from 10.0.0.1, port 47182
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 47184
[ ID] Interval      Transfer    Bitrate
[ 7] 0.00-1.00 sec  1.08 GBytes  9.26 Gbits/sec
[ 7] 1.00-2.01 sec  1.11 GBytes  9.48 Gbits/sec
[ 7] 2.01-3.01 sec  1.10 GBytes  9.48 Gbits/sec
[ 7] 3.01-4.00 sec  1.11 GBytes  9.60 Gbits/sec
[ 7] 4.00-5.00 sec  1.10 GBytes  9.48 Gbits/sec
[ 7] 5.00-6.00 sec  1.11 GBytes  9.52 Gbits/sec
[ 7] 6.00-7.01 sec  1.12 GBytes  9.60 Gbits/sec
[ 7] 7.01-8.00 sec  1.10 GBytes  9.51 Gbits/sec
[ 7] 8.00-9.00 sec  1.10 GBytes  9.41 Gbits/sec
[ 7] 9.00-10.00 sec 1.11 GBytes  9.59 Gbits/sec
[ 7] 10.00-10.02 sec 18.3 MBytes  9.17 Gbits/sec
[ ID] Interval      Transfer    Bitrate
[ 7] 0.00-10.02 sec 11.1 GBytes  9.49 Gbits/sec
Server listening on 5201
```

Рис. 10: Запуск iPerf3 на обоих хостах в разных режимах на обоих хостах

Интерактивные эксперименты

```
"host: h1"@mininet-vm

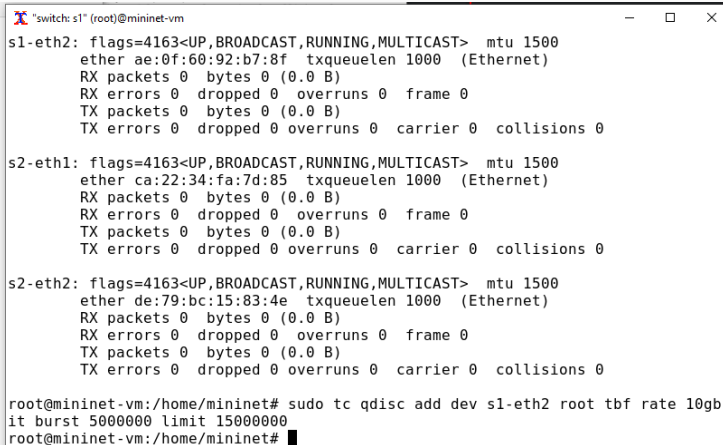
CONFIG_HZ_250=y
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
 7] local 10.0.0.1 port 47104 connected to 10.0.0.2 port 5201
ID] Interval      Transfer      Bitrate      Retr      Cwnd
7]  0.00-1.00    sec  1.09 GBytes  9.36 Gbits/sec      0      8.23 MBytes
7]  1.00-2.00    sec  1.11 GBytes  9.56 Gbits/sec      0      8.23 MBytes
7]  2.00-3.00    sec  1.10 GBytes  9.49 Gbits/sec      0      8.23 MBytes
7]  3.00-4.00    sec  1.11 GBytes  9.53 Gbits/sec      0      8.23 MBytes
7]  4.00-5.00    sec  1.11 GBytes  9.54 Gbits/sec      0      8.23 MBytes
7]  5.00-6.00    sec  1.12 GBytes  9.61 Gbits/sec      0      8.23 MBytes
7]  6.00-7.00    sec  1.10 GBytes  9.49 Gbits/sec      0      8.23 MBytes
7]  7.00-8.00    sec  1.09 GBytes  9.40 Gbits/sec      0      8.23 MBytes
7]  8.00-9.00    sec  1.12 GBytes  9.58 Gbits/sec      0      8.23 MBytes
7]  9.00-10.00   sec  1.11 GBytes  9.50 Gbits/sec      0      8.23 MBytes
- - - - -
ID] Interval      Transfer      Bitrate      Retr
7]  0.00-10.00   sec  11.1 GBytes  9.51 Gbits/sec      0
7]  0.00-10.02   sec  11.1 GBytes  9.49 Gbits/sec
sender
receiver

iperf Done.
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root
root@mininet-vm:/home/mininet#
```

Рис. 11: Удаление модифицированной конфигурации на хосте h1

При ограничении скорости на интерфейсе s1-eth2 коммутатора s1 все сеансы связи между коммутатором s1 и коммутатором s2 будут фильтроваться в соответствии с применяемыми правилами.

Интерактивные эксперименты



```
"switch: s1" (root)@mininet-vm
s1-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
        ether ae:0f:60:92:b7:8f txqueuelen 1000 (Ethernet)
        RX packets 0 bytes 0 (0.0 B)
        RX errors 0 dropped 0 overruns 0 frame 0
        TX packets 0 bytes 0 (0.0 B)
        TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

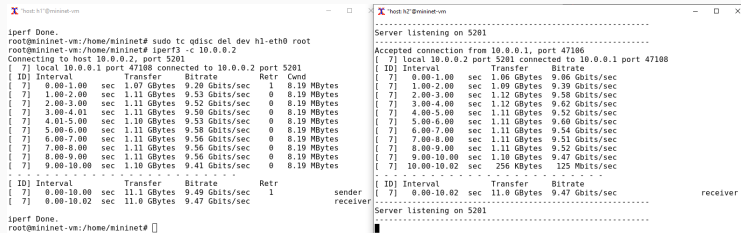
s2-eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
        ether ca:22:34:fa:7d:85 txqueuelen 1000 (Ethernet)
        RX packets 0 bytes 0 (0.0 B)
        RX errors 0 dropped 0 overruns 0 frame 0
        TX packets 0 bytes 0 (0.0 B)
        TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

s2-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
        ether de:79:bc:15:83:4e txqueuelen 1000 (Ethernet)
        RX packets 0 bytes 0 (0.0 B)
        RX errors 0 dropped 0 overruns 0 frame 0
        TX packets 0 bytes 0 (0.0 B)
        TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 root tbf rate 10gb
it burst 5000000 limit 15000000
root@mininet-vm:/home/mininet#
```

Рис. 12: Применение правила ограничения скорости

Интерактивные эксперименты

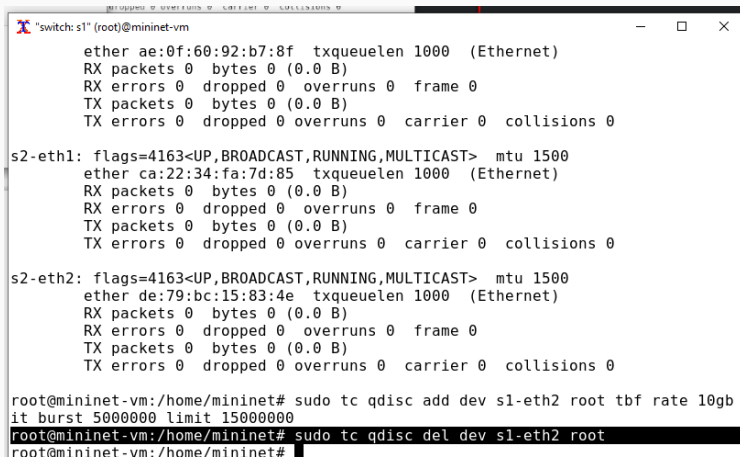


```
host: h1 @mininet-vm
iperf Done.
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 47188 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer      Bitrate      Retr  Cwnd
[ 7] 0.00-1.00 sec  1.07 GBytes  9.20 Gbits/sec  1    8.19 MBytes
[ 7] 1.00-2.00 sec  1.11 GBytes  9.53 Gbits/sec  0    8.19 MBytes
[ 7] 2.00-3.00 sec  1.11 GBytes  9.52 Gbits/sec  0    8.19 MBytes
[ 7] 3.00-4.01 sec  1.11 GBytes  9.50 Gbits/sec  0    8.19 MBytes
[ 7] 4.01-5.00 sec  1.10 GBytes  9.53 Gbits/sec  0    8.19 MBytes
[ 7] 5.00-6.00 sec  1.11 GBytes  9.58 Gbits/sec  0    8.19 MBytes
[ 7] 6.00-7.00 sec  1.11 GBytes  9.56 Gbits/sec  0    8.19 MBytes
[ 7] 7.00-8.00 sec  1.11 GBytes  9.56 Gbits/sec  0    8.19 MBytes
[ 7] 8.00-9.00 sec  1.11 GBytes  9.56 Gbits/sec  0    8.19 MBytes
[ 7] 9.00-10.00 sec 1.10 GBytes  9.41 Gbits/sec  0    8.19 MBytes
-----
[ ID] Interval      Transfer      Bitrate      Retr  sender receiver
[ 7] 0.00-10.00 sec 11.1 GBytes  9.49 Gbits/sec  1
[ 7] 0.00-10.02 sec 11.0 GBytes  9.47 Gbits/sec
iperf Done.
root@mininet-vm:/home/mininet#
```

```
host: h2 @mininet-vm
Server listening on 5201
Accepted connection from 10.0.0.1, port 47186
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 47186
[ ID] Interval      Transfer      Bitrate
[ 7] 0.00-1.00 sec  1.06 GBytes  9.06 Gbits/sec
[ 7] 1.00-2.00 sec  1.09 GBytes  9.39 Gbits/sec
[ 7] 2.00-3.00 sec  1.12 GBytes  9.58 Gbits/sec
[ 7] 3.00-4.00 sec  1.12 GBytes  9.62 Gbits/sec
[ 7] 4.00-5.00 sec  1.11 GBytes  9.52 Gbits/sec
[ 7] 5.00-6.00 sec  1.11 GBytes  9.60 Gbits/sec
[ 7] 6.00-7.00 sec  1.11 GBytes  9.54 Gbits/sec
[ 7] 7.00-8.00 sec  1.11 GBytes  9.51 Gbits/sec
[ 7] 8.00-9.00 sec  1.11 GBytes  9.52 Gbits/sec
[ 7] 9.00-10.00 sec 1.10 GBytes  9.47 Gbits/sec
[ 7] 10.00-10.02 sec 256 KBytes  125 Mbits/sec
-----
[ ID] Interval      Transfer      Bitrate
[ 7] 0.00-10.02 sec 11.0 GBytes  9.47 Gbits/sec
Server listening on 5201
```

Рис. 13: Запуск iPerf3 на обоих хостах в разных режимах на обоих хостах

Интерактивные эксперименты



```
"switch: s1" (root)@mininet-vm
    ether ae:0f:60:92:b7:8f txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

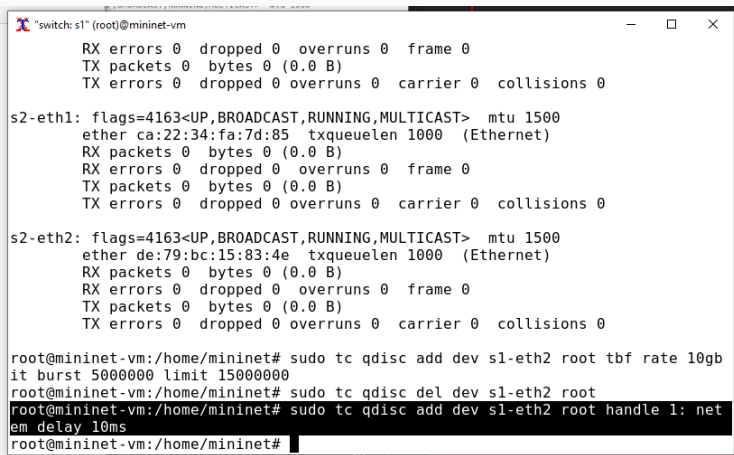
s2-eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether ca:22:34:fa:7d:85 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

s2-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether de:79:bc:15:83:4e txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 root tbf rate 10gb
it burst 5000000 limit 15000000
root@mininet-vm:/home/mininet# sudo tc qdisc del dev s1-eth2 root
root@mininet-vm:/home/mininet#
```

Рис. 14: Удаление модифицированной конфигурации на s1

NETEM используется для изменения задержки, джиттера, повреждения пакетов и т.д. TBF может использоваться для ограничения скорости. Утилита tc позволяет комбинировать несколько модулей. При этом первая дисциплина очереди (qdisc1) присоединяется к корневой метке, последующие дисциплины очереди можно прикрепить к своим родителям, указав правильную метку.



```
"switch: s1" (root)@mininet-vm
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

s2-eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
ether ca:22:34:fa:7d:85 txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

s2-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
ether de:79:bc:15:83:4e txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 root tbf rate 10gb
it burst 5000000 limit 15000000
root@mininet-vm:/home/mininet# sudo tc qdisc del dev s1-eth2 root
root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 root handle 1: net
em delay 10ms
root@mininet-vm:/home/mininet#
```

Рис. 15: Объединение NETEM и TBF

```
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 4
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
54 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=13.6 ms
54 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=11.2 ms
54 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=10.9 ms
54 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=10.3 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3008ms
rtt min/avg/max/mdev = 10.280/11.508/13.649/1.281 ms
root@mininet-vm:/home/mininet#
```

Рис. 16: Проверка соединения от хоста h1 к хосту h2

Интерактивные эксперименты

```
71      4.01 5.00      0.00 1.10 GBytes 0.53 Gbits/sec 0 0.10 MBytes
"switch: s1" (root)@mininet-vm
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

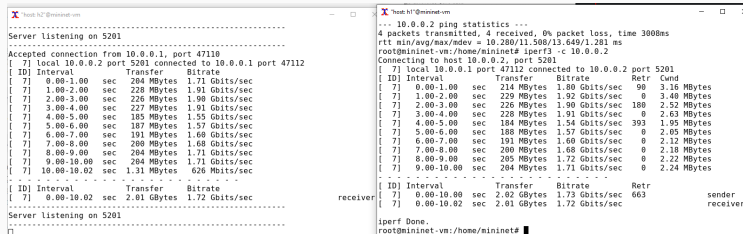
s2-eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
ether ca:22:34:fa:7d:85 txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

s2-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
ether de:79:bc:15:83:4e txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 root tbf rate 10gb
it burst 5000000 limit 15000000
root@mininet-vm:/home/mininet# sudo tc qdisc del dev s1-eth2 root
root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 root handle 1: net
em delay 10ms
root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 parent 1: handle 2
: tbf rate 2gbit burst 1000000 limit 2000000
root@mininet-vm:/home/mininet#
```

Рис. 17: Добавление второго правила

Интерактивные эксперименты



The image shows two terminal windows side-by-side. The left window, titled 'host h2'@mininet-vm, shows the server output of iPerf3. It reports an accepted connection from 10.0.0.1 on port 47112 and displays a table of performance metrics over 10-second intervals. The right window, titled 'host h1'@mininet-vm, shows the client output. It reports ping statistics (4 packets, 0% loss, 3008ms time) and then displays a similar table of performance metrics. Both tables show a transfer rate of approximately 1.71 Gb/s and a bitrate of 626 Mbits/sec.

```
-----
Server listening on 5201
-----
Accepted connection from 10.0.0.1, port 47112
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 47112
[ ID] Interval      Transfer    Bitrate
[ 7] 0.00-1.00 sec  204 MBytes  1.71 Gb/s/sec
[ 7] 1.00-2.00 sec  228 MBytes  1.91 Gb/s/sec
[ 7] 2.00-3.00 sec  226 MBytes  1.90 Gb/s/sec
[ 7] 3.00-4.00 sec  227 MBytes  1.91 Gb/s/sec
[ 7] 4.00-5.00 sec  185 MBytes  1.55 Gb/s/sec
[ 7] 5.00-6.00 sec  187 MBytes  1.57 Gb/s/sec
[ 7] 6.00-7.00 sec  191 MBytes  1.60 Gb/s/sec
[ 7] 7.00-8.00 sec  200 MBytes  1.68 Gb/s/sec
[ 7] 8.00-9.00 sec  204 MBytes  1.71 Gb/s/sec
[ 7] 9.00-10.00 sec 204 MBytes  1.71 Gb/s/sec
[ 7] 10.00-10.02 sec 1.31 MBytes  626 Mbits/sec
-----
[ ID] Interval      Transfer    Bitrate
[ 7] 0.00-10.02 sec 2.01 GBytes  1.72 Gb/s/sec
-----
Server listening on 5201
-----
```

```
--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3008ms
rtt min/avg/max/mdev = 10.280/11.508/13.649/1.281 ms
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 47112 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate      Retr      Cwnd
[ 7] 0.00-1.00 sec  214 MBytes  1.80 Gb/s/sec  90      3.16 MBytes
[ 7] 1.00-2.00 sec  229 MBytes  1.92 Gb/s/sec  0       3.40 MBytes
[ 7] 2.00-3.00 sec  226 MBytes  1.90 Gb/s/sec  180     2.52 MBytes
[ 7] 3.00-4.00 sec  228 MBytes  1.91 Gb/s/sec  0       2.63 MBytes
[ 7] 4.00-5.00 sec  184 MBytes  1.54 Gb/s/sec  393     1.95 MBytes
[ 7] 5.00-6.00 sec  188 MBytes  1.57 Gb/s/sec  0       2.05 MBytes
[ 7] 6.00-7.00 sec  191 MBytes  1.60 Gb/s/sec  0       2.12 MBytes
[ 7] 7.00-8.00 sec  200 MBytes  1.68 Gb/s/sec  0       2.18 MBytes
[ 7] 8.00-9.00 sec  205 MBytes  1.72 Gb/s/sec  0       2.22 MBytes
[ 7] 9.00-10.00 sec 204 MBytes  1.71 Gb/s/sec  0       2.24 MBytes
-----
[ ID] Interval      Transfer    Bitrate      Retr
[ 7] 0.00-10.00 sec 2.02 GBytes  1.73 Gb/s/sec  663
[ 7] 0.00-10.02 sec 2.01 GBytes  1.72 Gb/s/sec
-----
iperf Done.
root@mininet-vm:/home/mininet#
```

Рис. 18: Запуск iPerf3 на обоих хостах в разных режимах на обоих хостах

Интерактивные эксперименты

```
"switch: s1" (root)@mininet-vm

s2-eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether ca:22:34:fa:7d:85 txqueuelen 1000  (Ethernet)
    RX packets 0  bytes 0 (0.0 B)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 0  bytes 0 (0.0 B)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

s2-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether de:79:bc:15:83:4e txqueuelen 1000  (Ethernet)
    RX packets 0  bytes 0 (0.0 B)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 0  bytes 0 (0.0 B)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 root tbf rate 10gb
it burst 5000000 limit 15000000
root@mininet-vm:/home/mininet# sudo tc qdisc del dev s1-eth2 root
root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 root handle 1: net
em delay 10ms
root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 parent 1: handle 2
: tbf rate 2gbit burst 1000000 limit 2000000
root@mininet-vm:/home/mininet# sudo tc qdisc del dev s1-eth2 root
root@mininet-vm:/home/mininet#
```

Рис. 19: Удаление модифицированной конфигурации на s1

Самостоятельно реализуйте воспроизводимые эксперименты по использованию TBF для ограничения пропускной способности. Постройте соответствующие графики.

```
mininet@mininet-vm:~$ mkdir -p ~/work/lab_netem_iii/simple-tbf
mininet@mininet-vm:~$ cd ~/work/lab_netem_iii/simple-tbf
mininet@mininet-vm:~/work/lab_netem_iii/simple-tbf$ nano lab_netem_iii.py
mininet@mininet-vm:~/work/lab_netem_iii/simple-tbf$
```

Рис. 20: Создание каталога

```
GNU nano 4.8 lab_netem_iii.py
#!/usr/bin/env python
"""
Simple experiment.
Output: ping.dat
"""

from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI

from mininet.log import setLogLevel, info
import time

def emptyNet():
    """Create an empty network and add nodes to it."""
    net = Mininet( controller=Controller, waitConnected=True )

    info( '*** Adding controller\n' )
    net.addController( 'c0' )

    info( '*** Adding hosts\n' )
    h1 = net.addHost( 'h1', ip='10.0.0.1' )
    h2 = net.addHost( 'h2', ip='10.0.0.2' )

    info( '*** Adding switch\n' )
    s1 = net.addSwitch( 's1' )

    info( '*** Creating links\n' )
    net.addLink( h1, s1 )
    net.addLink( h2, s1 )

    info( '*** Starting network\n' )
    net.start()

    info( '*** Set rate\n' )

    h1.cmdPrint('tc qdisc add dev h1-eth0 root tbf rate 10gbit burst 5000000 limit 15000000')

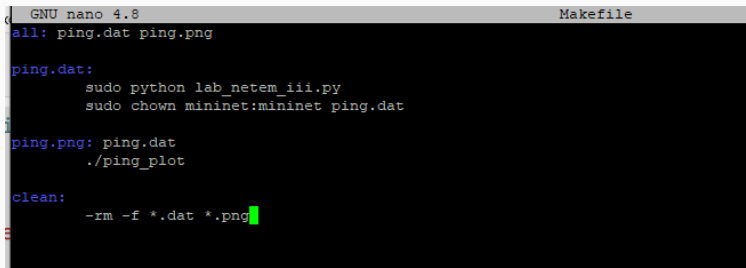
    time.sleep(10) # Wait 10 seconds

    info('*** Starting iperf3 server on h2\n')
    h2.cmdPrint('iperf3 -s 4')
    info('*** Running iperf3 client from h1 to h2\n')
    h1.cmdPrint('iperf3 -c ' + h2.IP() + ' | grep "MBytes" | awk '{print $7}' > ping.dat')

    info( '*** Stopping network' )
    net.stop()

if __name__ == '__main__':
    setLogLevel( 'info' )
    emptyNet()
```

Рис. 21: Скрипт lab_netem_iii.py

A screenshot of a terminal window showing the GNU nano 4.8 text editor editing a file named 'Makefile'. The editor's interface includes a title bar with 'GNU nano 4.8' on the left and 'Makefile' on the right. The main area shows the following Makefile content: 'all: ping.dat ping.png', 'ping.dat: sudo python lab_netem_iii.py sudo chown mininet:mininet ping.dat', 'ping.png: ping.dat ./ping_plot', and 'clean: -rm -f *.dat *.png'. The cursor is positioned at the end of the 'clean' rule. The background is black, and the text is in a light blue/cyan color.

```
GNU nano 4.8 Makefile
all: ping.dat ping.png

ping.dat:
    sudo python lab_netem_iii.py
    sudo chown mininet:mininet ping.dat

ping.png: ping.dat
    ./ping_plot

clean:
    -rm -f *.dat *.png
```

Рис. 22: Создание Makefile

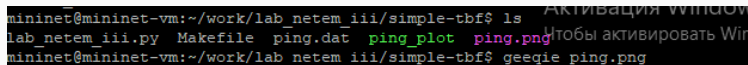
```
GNU nano 4.8                                     ping_plot
#!/usr/bin/gnuplot --persist
set terminal png crop
set output 'ping.png'
set xlabel "Packet number"
set ylabel "rate (Gbytes/sec)"
set grid
plot "ping.dat" with lines
```

Рис. 23: Создание ping_plot

Воспроизводимые эксперименты

```
mininet@mininet-vm:~/work/lab_netem_iii/simple-tbf$ make
sudo python lab_netem_iii.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Waiting for switches to connect
s1
*** Set rate
*** h1 : (tso qdisc add dev h1-eth0 root tbf rate 10gbit burst 5000000 limit 15000000,)
*** Starting iperf server on h2
*** h2 : ('iperf3 -s &')
*** Running iperf client from h1 to h2
*** h1 : ('iperf3 -c 10.0.0.2 | grep "MBytes" | awk '{print $7}' > ping.dat,)
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
...
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
sudo chown mininet:mininet ping.dat
./ping_plot
make: execvp: ./ping_plot: Permission denied
make: *** [Makefile:8: ping.png] Error 127
mininet@mininet-vm:~/work/lab_netem_iii/simple-tbf$ chmod +x ping_plot
mininet@mininet-vm:~/work/lab_netem_iii/simple-tbf$ make
./ping_plot
mininet@mininet-vm:~/work/lab_netem_iii/simple-tbf$ ls
lab_netem_iii.py Makefile ping.dat ping_plot ping.png
mininet@mininet-vm:~/work/lab_netem_iii/simple-tbf$
```

Рис. 24: Запуск скрипта и получение файлов



```
mininet@mininet-vm:~/work/lab_netem_iii/simple-tbf$ ls
lab_netem_iii.py  Makefile  ping.dat  ping_plot  ping.png
mininet@mininet-vm:~/work/lab_netem_iii/simple-tbf$ geeqie ping.png
```

The image shows a terminal window with a black background and white text. The first command is 'ls', which lists files in the current directory: 'lab_netem_iii.py', 'Makefile', 'ping.dat', 'ping_plot', and 'ping.png'. The second command is 'geeqie ping.png', which is used to open the image file. The prompt 'mininet@mininet-vm:' is visible at the start of each line. There is some faint, semi-transparent text on the right side of the terminal window, possibly from another window or a watermark.

Рис. 25: Запуск geeqie

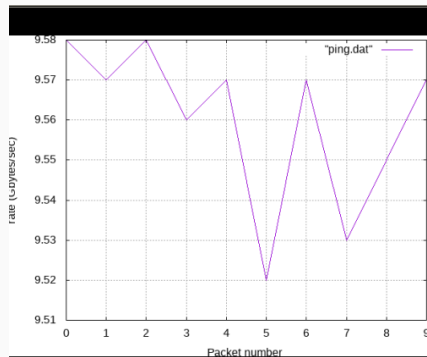


Рис. 26: Финальный график

Выводы

В результате выполнения данной лабораторной работы я познакомилась с принципами работы дисциплины очереди Token Bucket Filter, которая формирует входящий/исходящий трафик для ограничения пропускной способности, а также получила навыки моделирования и исследования поведения трафика посредством проведения интерактивного и воспроизводимого экспериментов в Mininet